

Mahmoud Zeidan

Faculty of Computer and Information Sciences (FCIS)
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PERSONAL DATA

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EXECUTIVE SUMMARY

Software Engineer & Computer Graphics Research Scientist with strong experience in scientific visualization, GPU programming, real-time rendering, and parallel computing. Proficient in C++, Python, and frameworks including CUDA, OpenGL, and Vulkan. Skilled in developing scalable visualization systems, geometric processing pipelines, and high-performance rendering architectures. Experienced in applying machine learning and deep learning techniques to enhance graphics, visualization, and computational performance, combining rigorous scientific research with robust engineering execution.

RESEARCH INTERESTS

Computer graphics, scientific visualization, GPU computing, real-time rendering, parallel processing, machine learning, and deep learning.

EXPERIENCE

Faculty of Computer and Information Sciences, Ain Shams University January 2026 – Present
Research Scientist and Lecturer Cairo, Egypt

Teaching Computer Graphics, Spring 2026.

Karlsruhe Institute of Technology (KIT) August 2016 – December 2025
Research and Teaching Assistant Karlsruhe, Germany

Developed visualization software and GPU-based frameworks for scientific data analysis at the Institute of Visualization and Data Analysis (IVD).

Assisted (2016–2020) in teaching GPGPU, GPU Computing, Graphics Programming, and Game Development.

Faculty of Computer and Information Sciences, Ain Shams University December 2005 – February 2016
Research and Teaching Assistant Cairo, Egypt

Conducted research on path tracing and photon mapping for realistic image synthesis.

Implemented optimized GPU rendering algorithms and assisted in teaching Mathematics, Physics, Probability & Statistics, and Structured Programming.

Microsoft Research ATLC October 2012 – March 2013
Research Assistant Intern Cairo, Egypt

Developed computer vision tools for object detection and image-based search.

Worked under the supervision of Dr. Motaz El-Saban.

Ractors Inc. May 2007 – December 2007
Web Developer Cairo, Egypt

Performed back-end web development for social media applications (HTML, JavaScript, PHP).

quTIP Software Company November 2006 – April 2007
Software Developer Cairo, Egypt

Developed desktop applications using C#, SQL Server, and .NET technologies.

EDUCATION

Karlsruhe Institute of Technology (KIT)

February 2016 – December 2025

PhD in Computer Science

Graduation: December 2025

Institute for Visualization and Data Analysis (IVD)

Thesis: *Glyph-based Display and Level-of-Detail Particle Rendering for Interactive Scientific Visualization*

Supervisor: Prof. Dr.-Ing. Carsten Dachsbacher

Ain Shams University, Egypt

June 2009 – July 2011

MSc in Computer Science

Thesis: Applying Parallel Processing Approach for Interactive Global Illumination

Supervisors: Prof. Taymour Nazmy, Prof. Mohamed Hashem, Dr. Haytham El-Messiry

Ain Shams University, Egypt

September 2000 – June 2004

BSc in Computer Science, Excellent with Honors

Thesis: "Photonix" – A 3D Modeling Tool Using Photon Mapping

General Secondary Education, Egypt

June 2000

Math Section, Grade: 98%

PUBLICATIONS

Mahmoud Zeidan. "Large Particle Datasets Visualization on GPUs." IVD - KIT, under submission.

Mahmoud Zeidan, Christoph Peters, Tobias Rapp, and Carsten Dachsbacher. "Versatile Geometric Flow Visualization by Controllable Shape and Volumetric Appearance." In Proceedings of Smart Tools and Applications in Graphics (STAG), 2022. [paper link](#).

Mahmoud Zeidan, Tobias Rapp, Christoph Peters, and Carsten Dachsbacher. "Moment-based Opacity Optimization." In Proceedings of Eurographics Symposium on Parallel Graphics and Visualization (EGPGV), 2020. [paper link](#).

Mahmoud Zeidan, Taymour Nazmy, and Mostafa Aref. "GPU-based Out-of-Core HLBVH Construction." In Proceedings of Eurographics Symposium on Rendering (EGSR) – Experimental Ideas & Implementation (EI&I), 2015. [paper link](#).

TECHNICAL STRENGTHS

Programming Languages:	C/C++, Python, Matlab — strong OOP design, clean and efficient code, SOLID principles.
GPU & Visualization:	OpenGL, GLSL, Vulkan, CUDA — advanced GPU programming, shader development, and performance tuning.
Database Systems:	Database modeling and design, SQL Server, MySQL, relational schema design.
Engineering Tools:	Git, CMake, Bash — workflow automation, CI, and cross-platform development.
Visualization & Design:	Blender, GIMP, Inkscape, Clip Studio Paint — research visualization and editing.
Operating Systems:	Linux (primary), Windows — system configuration and cross-platform workflows.

HOBBIES AND PERSONAL INTERESTS

Painting and modeling on physical and digital media, woodworking, nature walks, and writing and blogging in Arabic.

LANGUAGES

Arabic	: Native
English	: Professional working proficiency
German	: Proficient in daily communication